Stormwater Regulation Summary: Richmond, VA

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary drivers: James River watershed TMDL and Chesapeake Bay TMDL compliance.
- City of Richmond Department of Public Utilities administers local stormwater program.
- VESMP requirements apply + local Richmond Stormwater Utility requirements.
- Richmond has a combined sewer system in older neighborhoods — CSO reduction is a priority.
- Richmond 300 Master Plan encourages green infrastructure and sustainable development practices.

2. Typical Soil Conditions

- Richmond straddles the Fall Line — Piedmont clays to the west, Coastal Plain sands to the east.
- West of Fall Line (Fan, Museum District, Church Hill): residual Piedmont clay; HSG C-D.
- East of Fall Line (Shockoe Bottom, East End): Coastal Plain sandy loam and alluvial deposits; HSG B-C.
- James River floodplain soils are alluvial — variable drainage, moderate to good infiltration where not compacted.
- Implication: Site location relative to the Fall Line significantly affects infiltration feasibility — test before designing.
- *Historic fill is common in developed areas — soil investigation required regardless of mapped soil type.*

3. Green Roof Requirements

- No green roof mandate in Richmond.
- Green roofs receive phosphorus removal credit under VESMP — effective for meeting Virginia's pollutant-based requirements.
- Richmond stormwater utility fee credits available for on-site stormwater management, including green roofs.
- VCU and VCU Health System have institutional sustainability commitments that support green roof adoption.
- Scott's Addition and Manchester redevelopment projects face tight site constraints — rooftop BMPs are practical.
- Richmond 300 plan identifies green infrastructure as a priority strategy for urban resilience.

4. TSS Requirements

- Richmond follows Virginia VESMP phosphorus-based design approach — BMPs credited for TP removal.
- TSS removal is a co-benefit of phosphorus-focused BMPs — green roofs achieve both TP and TSS reduction.
- James River watershed TMDL establishes nutrient reduction targets that green infrastructure helps address.
- Combined sewer areas benefit from any volume reduction — green roofs reduce both CSO frequency and pollutant loading.
- *Confirm current pollutant removal credits for green roofs in Virginia BMP Clearinghouse.*

5. Nature-Based Retention Requirements

- VESMP phosphorus-based design requirements apply — retention volume that achieves required TP load reduction.
- Practical equivalent: approximately 1.0" retention standard for most Richmond sites.
- CN (Curve Number) method is the standard design approach.
- Richmond DPU reviews stormwater management plans for compliance with both local and state requirements.
- Combined sewer system areas: Any retention reduces CSO discharge to James River — additional regulatory value.
- *Richmond may have local amendments to VESMP standards — verify with DPU.*

6. Allowable Outflow Rates

- VESMP standards apply: 24-hr extended detention of 1-yr storm (Cpv) + 10-yr flood protection.
- Richmond does not specify a flat l/s/ha rate — control is relative to pre-development conditions.
- Combined sewer areas may have additional discharge restrictions based on sewer capacity.
- For reference, typical pre-development Richmond site (CN 65-70): approximately 15-30 l/s/ha for the 10-year storm.
- *Specific allowable rates depend on sewer shed — Richmond DPU approval required for each connection.*